The Personal Income Tax: In-Class Questions

EC313 Public Economics: Taxation, Fall 2026

These questions are written in the style of the short-answer section of the midterm, where each part is worth 5 marks. We work through the answers in class, so solutions are not posted.

Question 1

Priya bought shares for $10,000 that are now worth $20,000. Half of a capital gain is taxable, and her tax rate is 40%. She will sell in one year no matter what she decides now. Her shares will earn 10% over the year; another stock will earn 11%.

a) If she sells her shares now to buy the other stock, how much tax does she pay today, and how much can she invest in the other stock?

b) Should she switch? Use your answer to explain the lock-in effect.

Question 2

For each of the following, state whether it is true or false, and provide an explanation (all marks come from the explanation).

a) Under the Haig-Simons definition, an increase in the value of shares you still own is not income until you sell them.

b) Taxing a capital gain when the asset is sold is equivalent to taxing it each year as it accrues, as long as the tax rate is the same.

Question 3

A province introduces a refundable work credit equal to 25% of earnings above $4,000, up to a maximum of $2,000. The credit is reduced by 20% of earnings above $30,000 until it is gone. The province's only other tax is a 20% income tax on earnings above $15,000. Ignore all other taxes and benefits.

a) At what earnings does the credit reach its maximum, and at what earnings is it gone?

b) Find the effective marginal tax rate for each range of earnings from $0 to $50,000. What does this schedule do to the incentive to take a job, and to the incentive for a part-time worker earning $30,000 to work more hours?